

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Ethical Analysis Report

Group ID: 25-26J-090

Project Title:

AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT MANAGEMENT IN STREAMING ENVIRONMENTS

	Identification and Analysis
Data Privacy Concerns	The framework processes real-time streaming data that may contain personal or sensitive attributes. The Privacy Preservation service identifies and masks such fields at the point of ingestion, before data is written to any storage layer. This reduces the risk of exposing personal information during transit and supports compliance with data minimization principles. Sensitive attribute detection is performed at the attribute level, ensuring that protective actions are applied precisely without unnecessarily discarding useful analytical information. The modular design allows privacy protection to be adopted as a standalone component, enabling organizations in healthcare, finance, and other regulated sectors to integrate it into existing pipelines without overhauling their full architecture.
Data Security	The framework incorporates a staging queue mechanism for the Schema Drift Detection service, which routes records with high-risk schema changes away from production systems pending human review. This prevents unvalidated or malformed data from propagating downstream and disrupting operational workflows. The Privacy Preservation service applies masking and transformation actions in transit, ensuring that sensitive fields never reach storage in their raw form. Together, these design choices reduce the attack surface for unauthorized access to sensitive data and limit the consequences of a potential breach by ensuring that stored data has already been cleaned and protected at the pipeline layer.
Bias and Fairness	The Structured Data Cleaning service uses statistical methods and machine learning models to detect missing values, outliers, and invalid records. There is a risk that these models may encode systematic biases if they are trained on unrepresentative historical data, potentially causing certain types of records to be disproportionately flagged or corrected. To mitigate this, the framework incorporates a user-in-the-loop review step, which allows data engineers to inspect and override suggested corrections before they are applied. Automated fallback fills are used only for records where no manual decision has been made, limiting the scope of unchecked automated choices. The risk-scoring mechanism in the Schema Drift Detection service also requires transparency in how severity is assigned to avoid unintended bias in routing decisions.

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Transparency and Accountability	The framework is designed with explainability and auditability at its core. Each service produces traceable outputs: schema drift severity scores with the reason for flagging, identified privacy actions per attribute, quality correction suggestions with the option for human override, and preprocessing logs for unstructured data. The user-in-the-loop design ensures that humans remain accountable for consequential decisions, particularly in data cleaning and schema routing. The modular architecture means that each service can be documented, tested, and audited independently. Clear separation of concerns across the four services makes it straightforward for teams to understand which component made a specific decision, supporting accountability for ethical implications throughout the pipeline lifecycle.
Impact on Stakeholders	Data engineering and AI teams benefit from more reliable pipeline outputs and reduced time diagnosing silent failures. Organizations in healthcare, finance, and education benefit from stronger privacy guarantees before data reaches storage. End users whose data flows through connected systems benefit indirectly from consistent in-transit protection. Care must be taken to ensure that automated decisions, such as outlier removal or schema rerouting, do not disproportionately affect legitimate records. The user-in-the-loop mechanism serves as a safeguard, enabling human judgment to override system actions where a more nuanced approach is required.